

Project Design Phase

Problem Architecture

Date	30 September 2026
Team ID	SWTID-2026-1617
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	4 Marks

Problem Architecture:

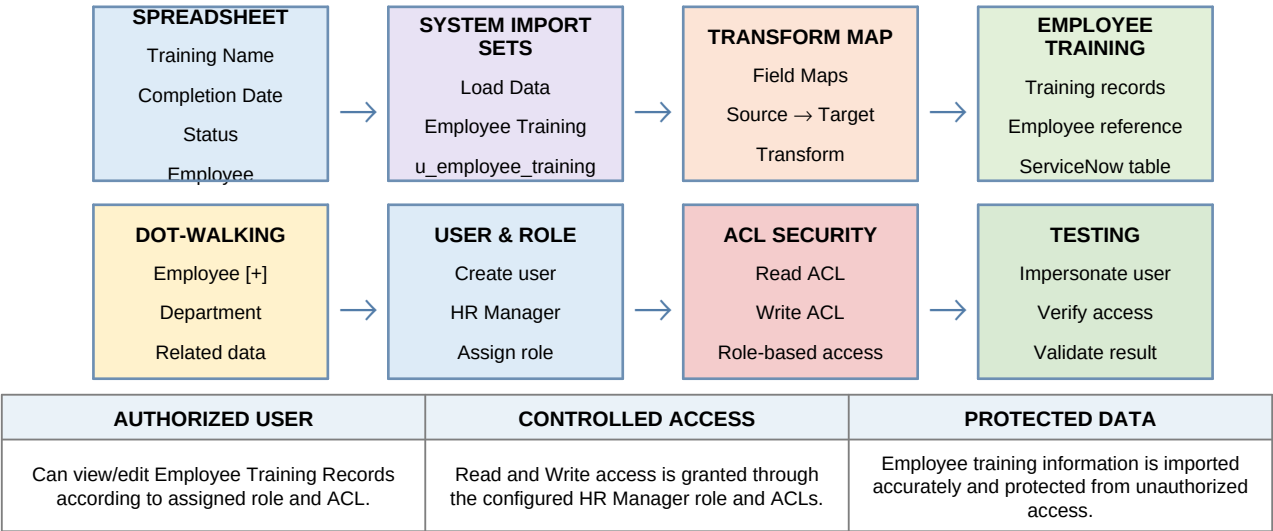
The architecture addresses the need to import employee training data from a spreadsheet into ServiceNow, connect each training record to an employee, access related Department information through dot-walking, and protect the resulting records with role-based access control and ACLs. The source project uses Employee Training Records with Training Name, Completion Date, Status and Employee fields.

Architecture Components:

- **Data Source:** Spreadsheet containing Training Name, Completion Date, Status and Employee.
- **Import Layer:** System Import Sets → Load Data uploads the spreadsheet and creates the Employee Training import data.
- **Transformation Layer:** Transform Maps and Field Maps map source fields to the Employee Training Records table.
- **Relationship Layer:** Employee is a reference field to sys_user; dot-walking exposes Department information.
- **Security Layer:** Users and the HR Manager role are configured, followed by Read and Write ACLs for Employee Training Records.
- **Validation:** Impersonation is used to verify that an authorized user can view/edit records and another user cannot access the table.

Problem Architecture Diagram:

Importing & Securing Data in ServiceNow — Data and Access Flow



Architecture Rule: Spreadsheet data is loaded through System Import Sets, mapped with Transform Maps, connected to employee-related data using dot-walking, and secured through users, roles and ACLs. Testing with impersonation confirms the configured access behavior.

Figure 1: Architecture and data-security flow of the Importing & Securing Data in ServiceNow solution